

Computer Science & IT

Database Management System

Entity Relationship Model
&
Integrity constraints

Lecture No. 03



By- Vishal Sir



Recap of Previous Lecture



Topic

ER model & ER diagram



Topic

Integrity constraints



Topics to be Covered



Topic

Integrity constraints



Topic

Conversion from ER model to relational table





Topic : Relational Model concept

- ❑ The relational model was proposed by E.F. Codd to model data in the form of relations or tables.
- ❑ The relational model is represented by a table with columns and rows. Each row is known as a tuple.



Topic : Relational Constraints

- These are the restrictions or set of rules imposed on the database content. It validates the quality of the database. It validates that various operations like data insertion, updation, etc., can be performed without affecting the integrity of the data.

Constraints could be either on a column level or a table level. The column level constraints are applied only to one column, whereas the table level constraints are applied to the whole table.



Topic : Types of Constraints on Relational Model

- ✓ 1. Domain constraint
- ✓ 2. Key constraint or Tuple Uniqueness constraint
- ✓ 3. Entity Integrity constraint
- ④ 4. Referential Integrity constraint



Topic : Domain constraint

Every domain must contain atomic values(smallest indivisible units) which means composite and multi-valued attributes are not allowed.

We perform a datatype check here, which means when we assign a data type to a column we limit the values that it can contain. Eg. If we assign the datatype of attribute age as int, we can't give it values other than int datatype



Topic : Key constraint or Tuple Uniqueness constraint

- These are called uniqueness constraints since it ensures that every tuple in the relation should be unique.
it states that Every relation must have a key, and for that tuples in the relation must be distinct
- A relation can have multiple keys or candidate keys(minimal superkey), out of which we choose one of the keys as the primary key, we don't have any restriction on choosing the primary key out of candidate keys.



Topic : Entity Integrity constraint

- Entity integrity constraint specifies that no attribute of primary key must contain a null value in any relation.



Topic : Referential Integrity constraint

MIMP



The Referential integrity constraint is specified between two relations or tables and used to maintain the consistency among the tuples in two relations.

Some times
may be on the
same relation



Topic : Referential Integrity constraint

Foreign Key: A foreign key is a set of attributes in a table that refers to the
primary key or alternate key of same table or some other table.

eg:

Names of the attributes need not be same

Foreign key

it references to the "Stu-id" attribute of Student relation

Primary

Key of student

<u>Sid</u>	<u>Cid</u>
S ₁	C ₁
S ₁	C ₂
S ₃	C ₂
S ₄	C ₃

Enroll

Enroll relation is the "referencing" relation

it is also a foreign key which will reference the course relation

<u>Stu-id</u>	Sname	Branch
S ₁		
S ₂		
S ₃		
S ₄		

Student

Student relation is the "referenced" relation

<u>Cid</u>	Cname	Branch
C ₁		
C ₂		
C ₃		
C ₄		

Course

Note:

① A relation that contains foreign key is called a referencing relation

② Relation that contains the primary key / alternate key which is being referenced by the foreign key is called a referenced relation

in some cases,

referencing relation

referenced relation may be same

that references
to the primary
key of
same table

Primary key

it is
foreign key

<u>Eid</u>	Ename	...	Manager-id
E1			E3
E2			E5
E3			E6
E4			E6
E5			NULL
E6			E5

Employee

Employee relation is
referencing relation
as well as
referenced relation

Foreign key attributes
may take NULL value



Topic : Referential Integrity constraint

- Referential integrity constraint is enforced through a foreign key
- Let foreign key in relation R1 refers to primary key of relation R2.

The values of the foreign key in a tuple of relation R1 can either take the values of the primary key ~~from~~ some tuple in relation R2, or can take NULL values, but can't be empty.

$$\left\{ \begin{array}{l} \text{Values in the} \\ \text{Foreign Key} \\ \text{Column} \end{array} \right\} = \left\{ \begin{array}{l} \text{Values in the} \\ \text{P.K. / alternate} \\ \text{Column} \end{array} \right\} \cup \{ \text{NULL} \}$$



Topic : Referential Integrity constraints on Referenced relation



Insertion: Insertion of a tuple in referenced relation does not result in any referential integrity violation.

Deletion: Deletion of a tuple from referenced relation may result in referential integrity violation.

Updation: Updation of a tuple in referenced relation may result in referential integrity violation.



Topic : Referential Integrity constraints on Referencing relation

Insertion: Insertion of a tuple in referencing relation may result in referential integrity violation.

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Updation: Updation of a tuple in referencing relation may result in referential integrity violation.



Topic : Referential Integrity constraints on Referenced relation

(or Alternate key value)

Because of deletion

- ✦ Deletion of a primary key value¹ from the referenced relation may causes integrity violation. We may choose one of the following approaches to avoid integrity violation.

✓ On Delete No Action:

✓ On Delete Cascade:

✓ On Delete Set NULL:

On delete No action

If deletion of a tuple from referenced relation causes any integrity violation, then that deletion is prohibited.

{ i.e. No action will be performed }

Not even the deletion from referenced relation

On delete Cascade

If deletion of a tuple from referenced relation causes any integrity violation, then delete the tuple from the referenced relation, and also delete the corresponding tuples from referencing relation.

it may result in deletion of few more tuples, if referencing relation & referenced relation is same

On delete Set NULL

Case ① If deletion of a tuple from referenced relation causes any integrity violation and foreign key attribute is allowed to take the NULL value, then delete the tuple from the referenced relation and set the value of the foreign key attribute as "NULL" in the tuples that causes integrity violation.

Case ② If deletion of a tuple from referenced relation causes any integrity violation and foreign key attribute is not allowed to take the NULL value, then deletion from the referenced relation itself is prohibited, i.e., if foreign key attribute is not allowed to take NULL values then "On delete Set NULL" is same as "On delete no action".



Topic : Referential Integrity constraints on Referenced relation

Because of updation

- Updation in the primary key value in the referenced relation may causes integrity violation. We may choose one of the following approaches to avoid integrity violation.

On Update No Action:

On Update Cascade:

On Update Set NULL:

On Update No action

If updation of a tuple in referenced relation causes any integrity violation, then that updation is prohibited.

{ i.e. No action will be performed }

Not even the updation in referenced relation

On Update Cascade

If updation of a tuple in referenced relation causes any integrity violation, then update the tuple in the referenced relation, and also update the values in the foreign key column of the referencing relation.

On Update Set NULL

Case ① If updation of a tuple in referenced relation causes any integrity violation and foreign key attribute is allowed to take the NULL value, then update the tuple in the referenced relation and set the value of the foreign key attribute as "NULL" in the tuples that causes integrity violation.

Case ② If updation of a tuple in referenced relation causes any integrity violation and foreign key attribute is not allowed to take the NULL value, then updation in the referenced relation itself is prohibited, i.e., if foreign key attribute is not allowed to take NULL values then "On Update Set NULL" is same as "On Update No action".



Topic : Referential Integrity constraints on Referencing relation



★ **Insertion:** Insertion of a tuple in referencing relation may result in referential integrity violation.

Deletion: Deletion of a tuple from referencing relation does not result in any referential integrity violation.

Updation: Updation of a tuple in referencing relation may result in referential integrity violation.



Topic : Referential Integrity constraints on Referencing relation



If any operation in referencing relation (child relation) causes any referential integrity violation, then corresponding operation is prohibited.

ER Model to Relational Table

Objective:-

Given an
ER diagram



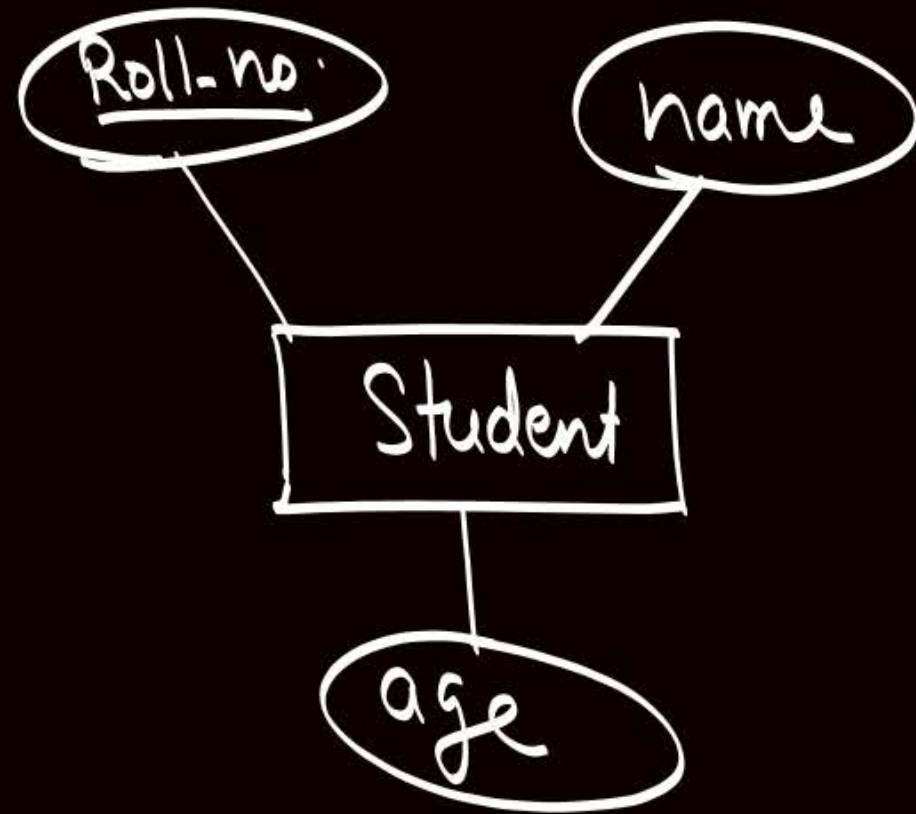
We need to
Convert it
in to



Relational
Model
{ relational table }

★ Given an entity set, we will define relation w.r.t. that entity set.

① All attributes are Simple attributes: -



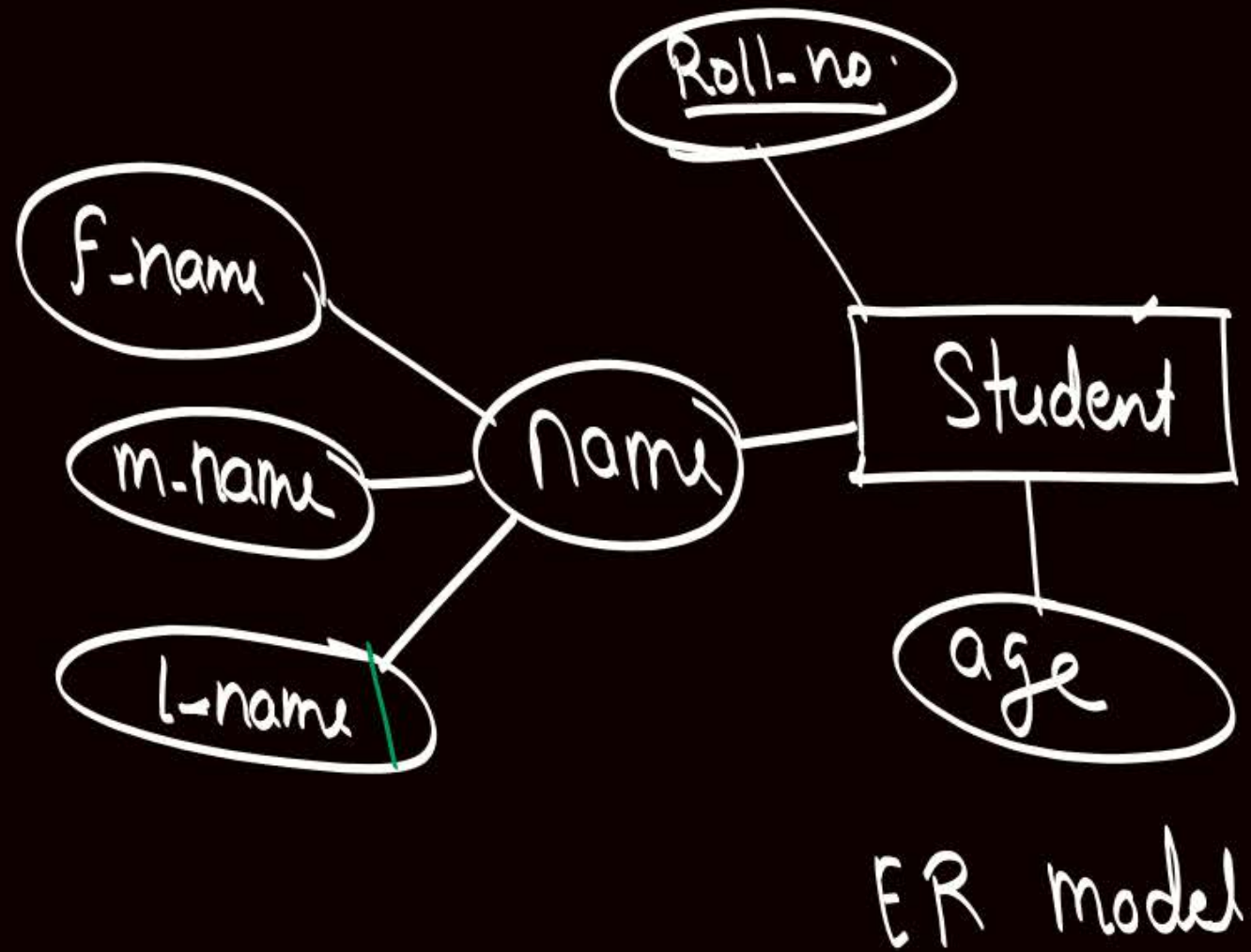
ER model

→
Relational
table

Student

<u>Roll-no</u>	name	age

- ★ Given an entity set, we will define relation w.r.t. that entity set.
- ② When some attributes are Composite attributes:-



Relational table

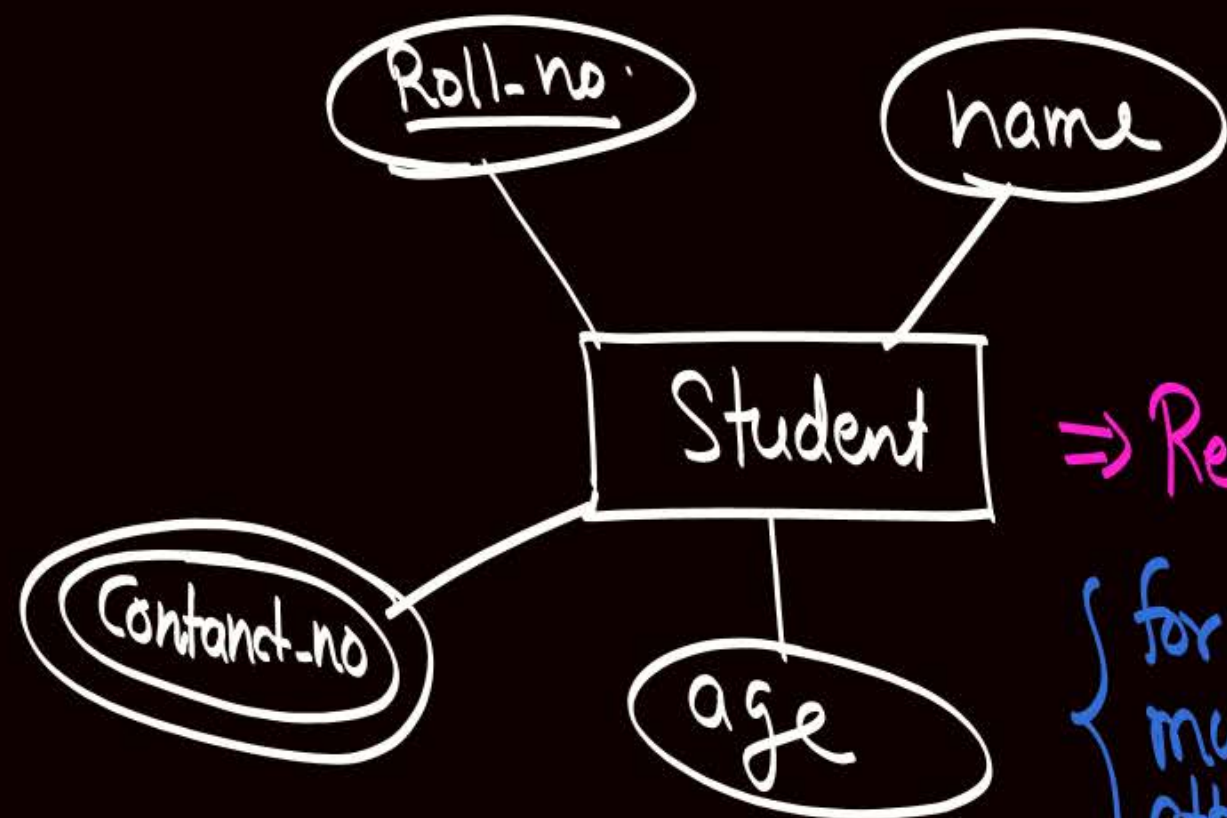
In the relational table, we will consider only simple components of composite attribute

Student

<u>Roll-no</u>	f-name	m-name	l-name	Age

* Given an entity set, we will define relation w.r.t. that entity set.

③ Some attributes are multivalued attributes:



ER model

⇒ Relational tables ⇒ Two relational tables

{ for each mult.-valued attribute, we will require a separate relation }

One, w.r.t. all simple attributes

one w.r.t. multivalued attribute "Contact-no"

Student

<u>Roll-no.</u>	name	age

Stu-Contact-info

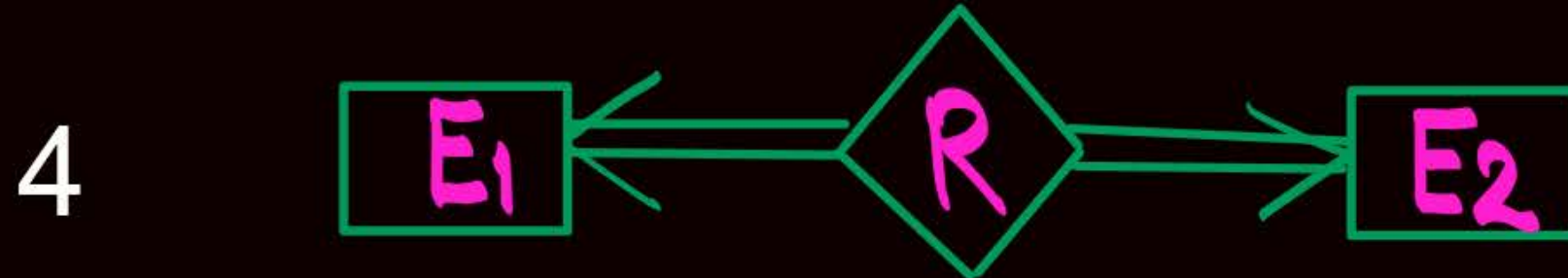
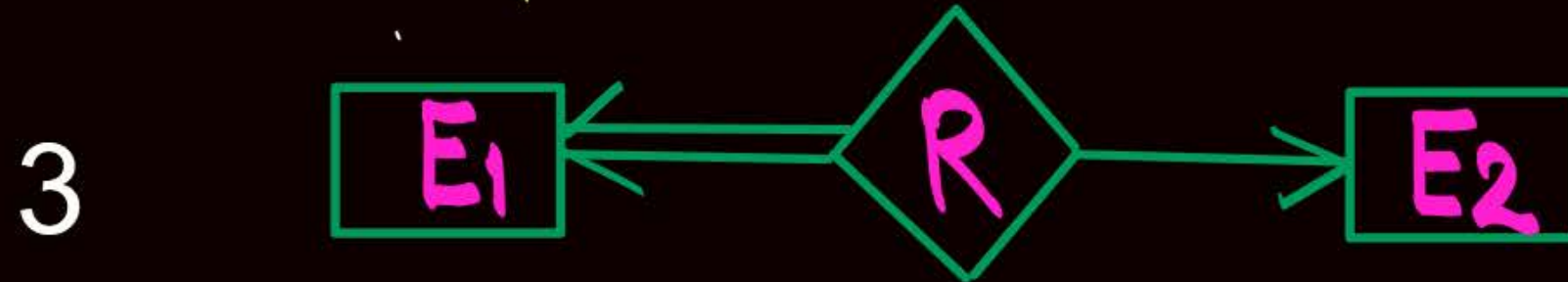
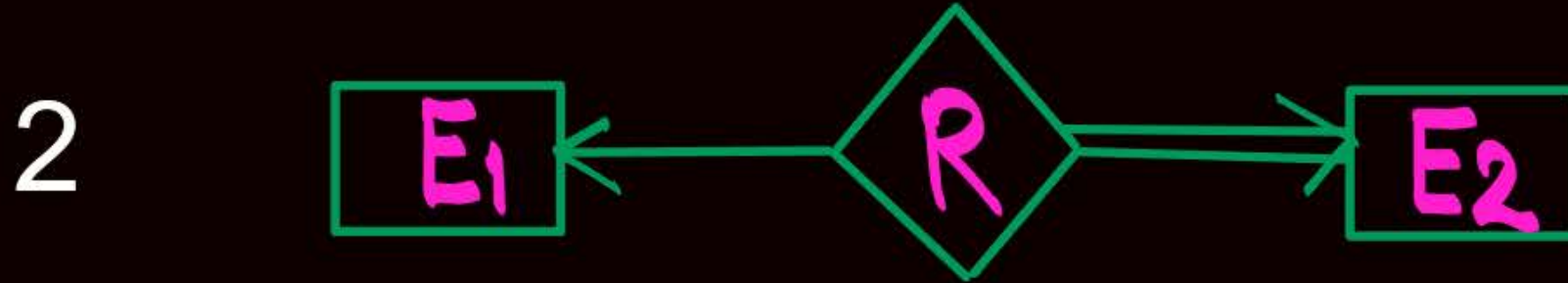
<u>Roll-No.</u>	<u>Contact-no</u>
101	94278
101	99930
102	98270

Note:

There is no representation of Composite attributes and multivalued attributes in relational model

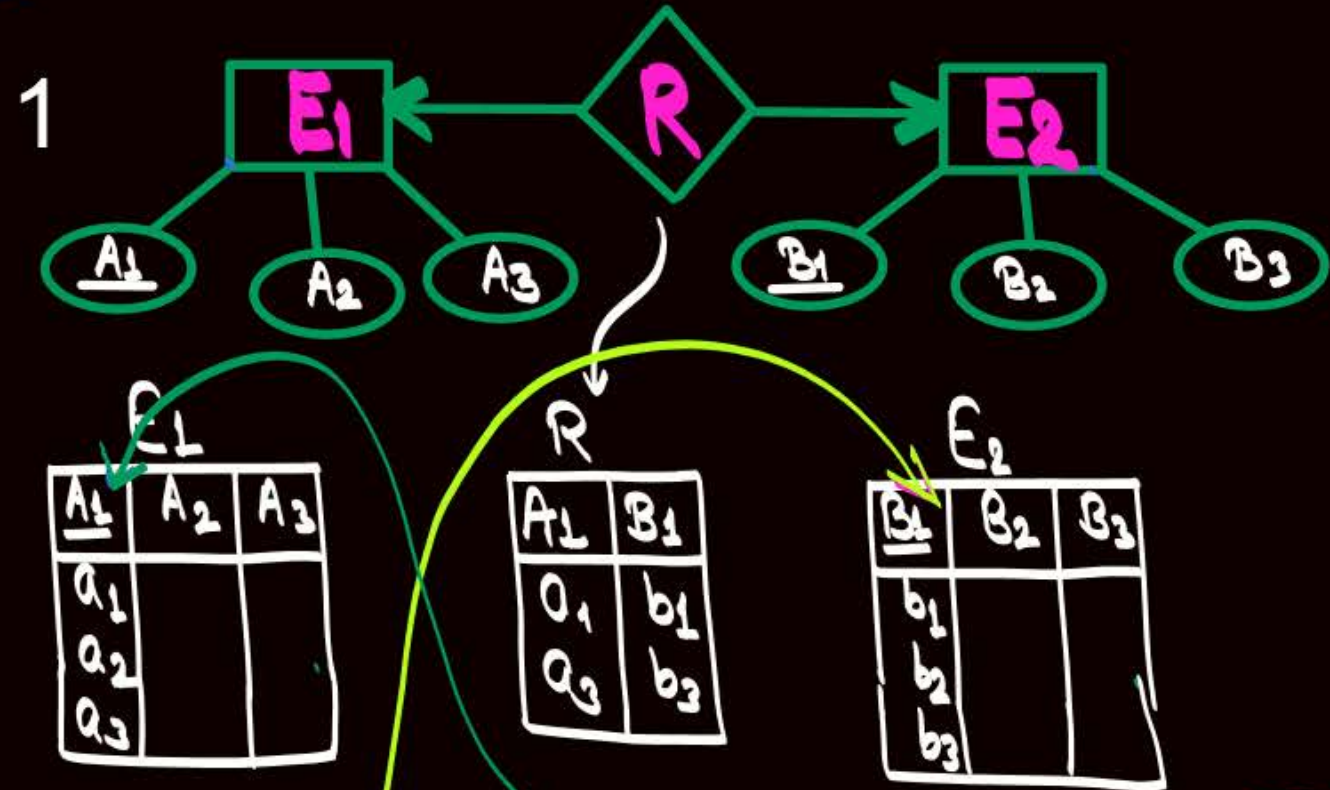
ER Model to Relational Table

① One to One Relation



ER Model to Relational Table

① One to One Relation { Partial participation of both side entities }



$E_1 E_2 R$

A_1	A_2	A_3	B_1	B_2	B_3
a_1	-	-	b_1	-	-
a_2	-	-	NULL	NULL	NULL
a_3	-	-	b_3	-	-
NULL	NULL	NULL	b_2	-	-

Can not be combined into a single relation, because primary key can not be defined

P.K. $E_1 R$ F.K.

<u>A_1</u>	A_2	A_3	B_1
a_1	-	-	b_1
a_2	-	-	NULL
a_3	-	-	b_3

R.K. $E_2 R$ F.K.

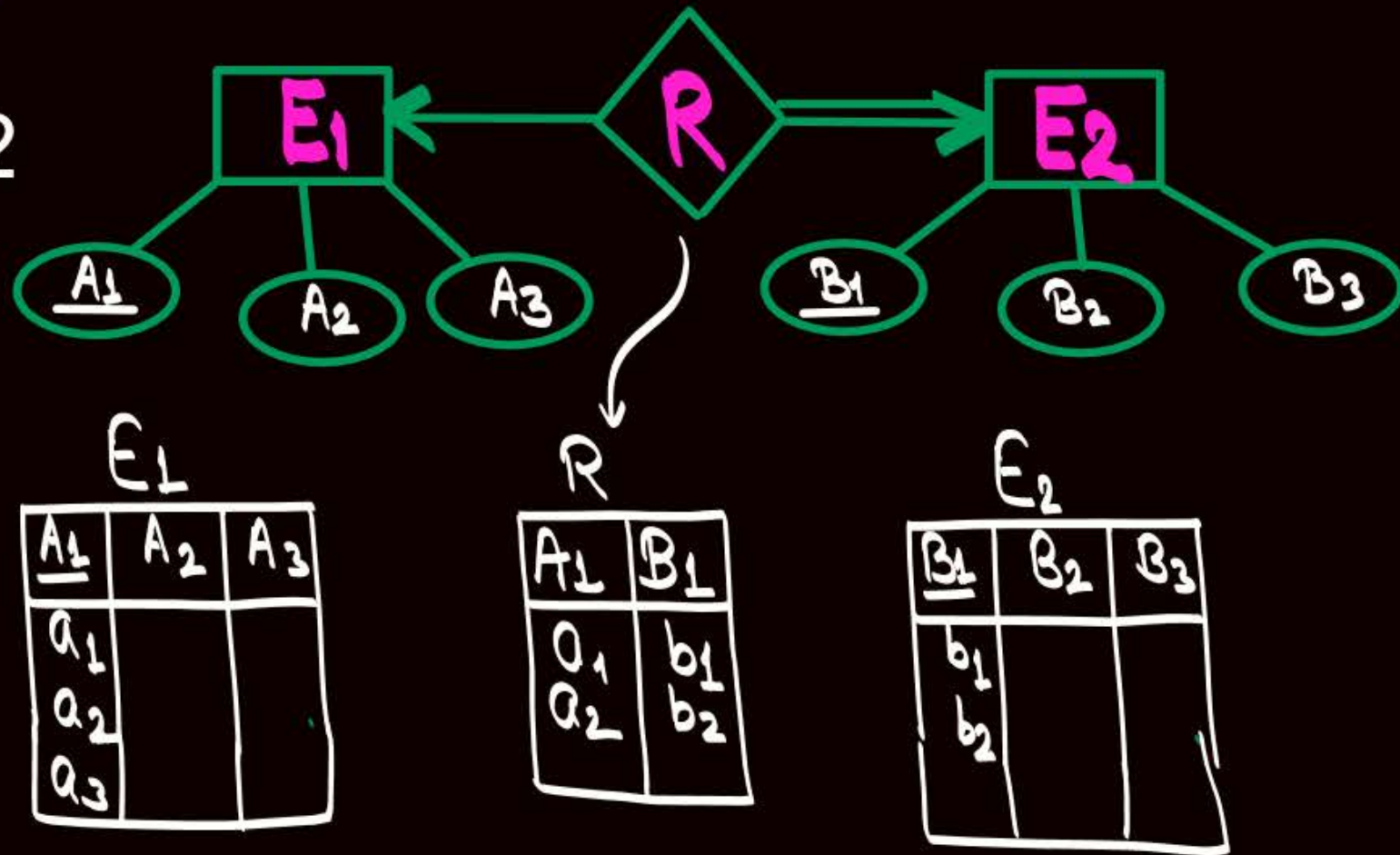
(or)

<u>B_1</u>	B_2	B_3	A_1
b_1	-	-	a_1
b_2	-	-	NULL
b_3	-	-	a_3

ER Model to Relational Table

① One to One Relation { total Participation of one entity }

2



Primary Key (E1 E2 R)

<u>A1</u>	A2	A3	B1	B2	B3
a1	-	-	b1	-	-
a2	-	-	b2	-	-
a3	-	-	NULL	-	-

P.K = A1

Same as P.K = E1
[Entity with Partial Participation]

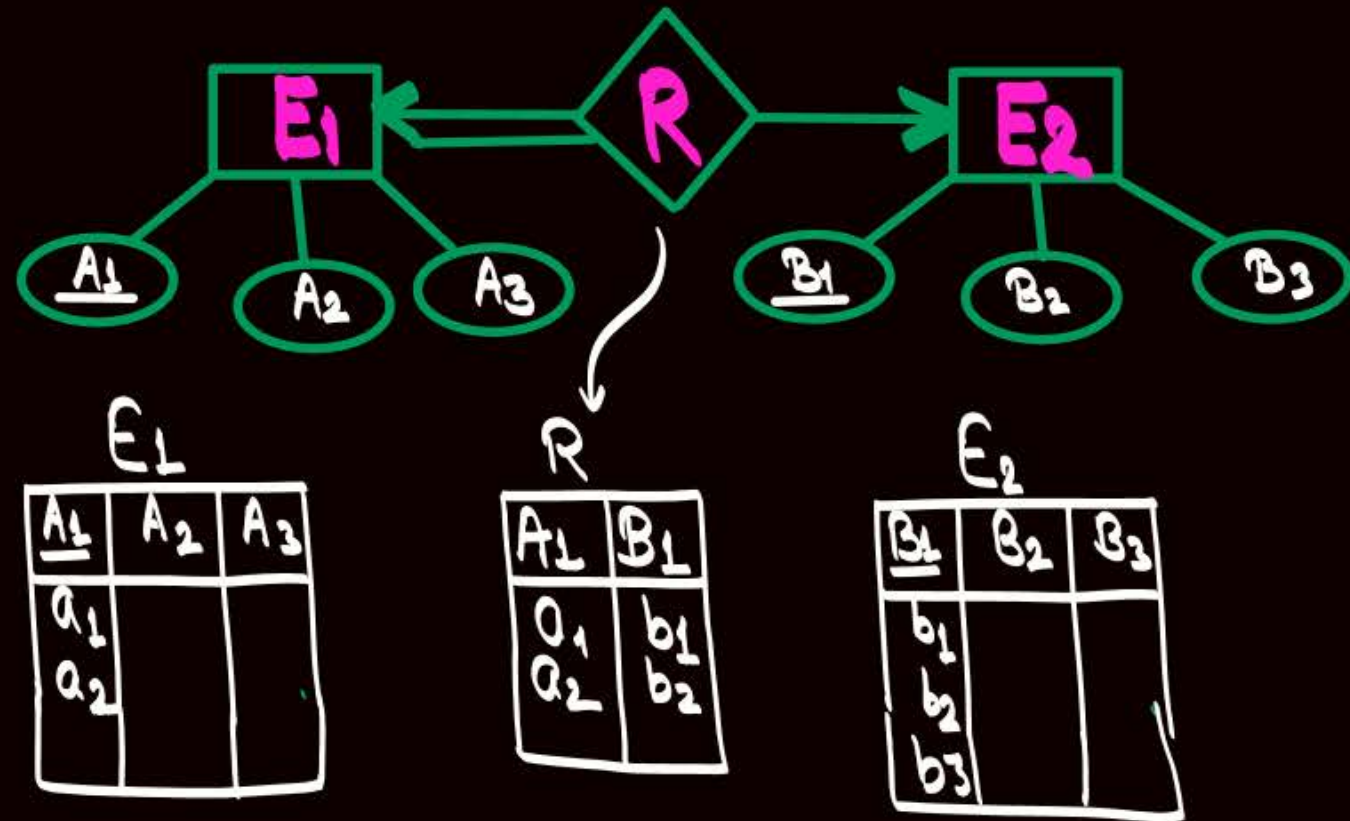
We can combine all three table into a single relation.

Primary Key of the merged relation will be same as Primary Key of the Entity with Partial Participation

ER Model to Relational Table

① One to One Relation { total Participation of one entity }

3



it is same as previous case,
all three can be combined
into a single table.

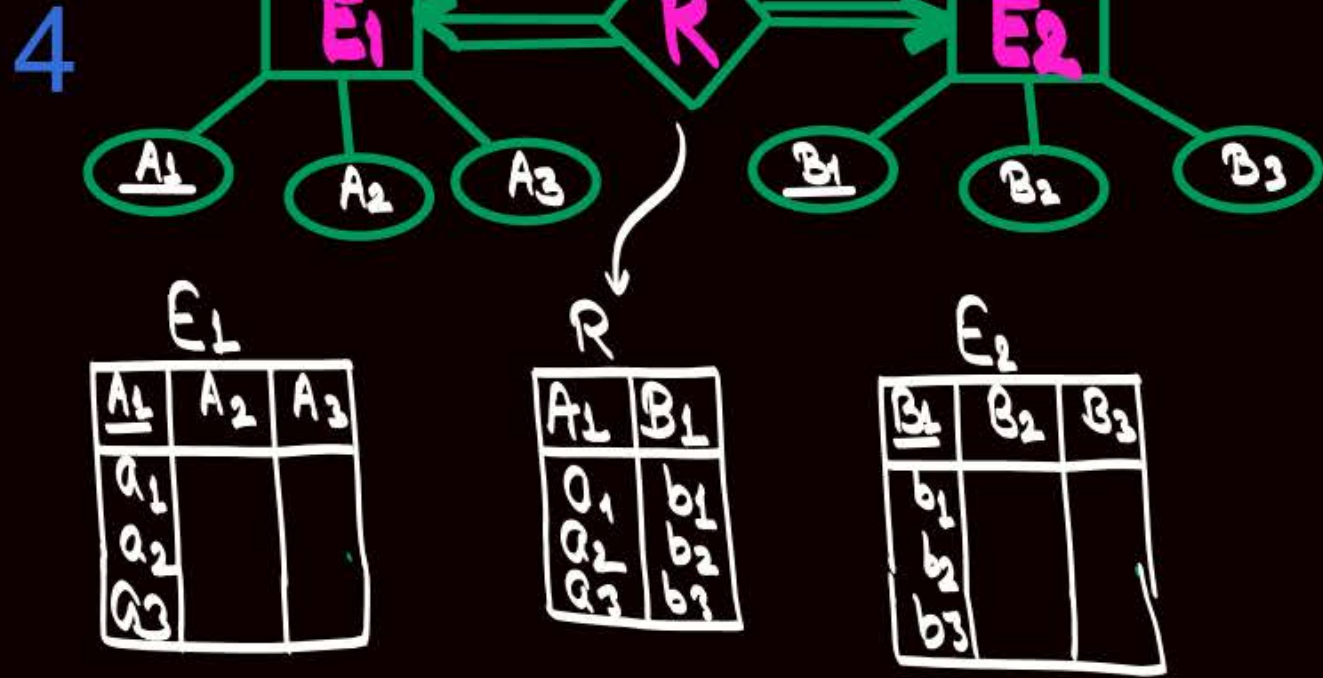
Primary Key of
merged relation

= (B1)

ie, Primary key of the
Entity set with
Partial participation

ER Model to Relational Table

① One to One Relation { total participation of both side entity set }



$E_1 E_2 R$					
A_1	A_2	A_3	B_1	B_2	B_3
a_1	-	-	b_1	-	-
a_2	-	-	b_2	-	-
a_3	-	-	b_3	-	-

Primary Can be
(A_1) or (B_1)

All three can be merged into a single relation, and primary key of merged relation can be primary key of either side of entity set

ER Model to Relational Table

① One to One Relation

1



{ minimum two tables are required }

2



{ One table is required
{ P.K. of Combined relation will be same }
as P.K. of the entity set with Partial participation }

3



{ One table is required
{ P.K. of Combined relation will be same }
as P.K. of the entity set with Partial participation }

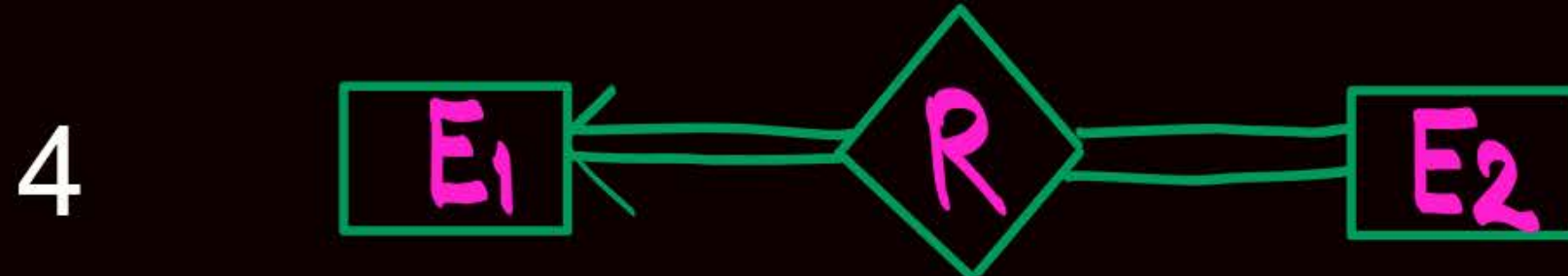
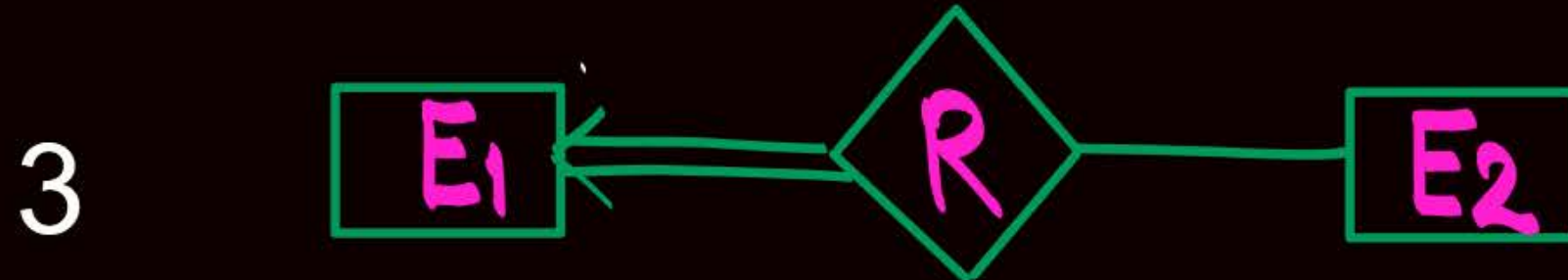
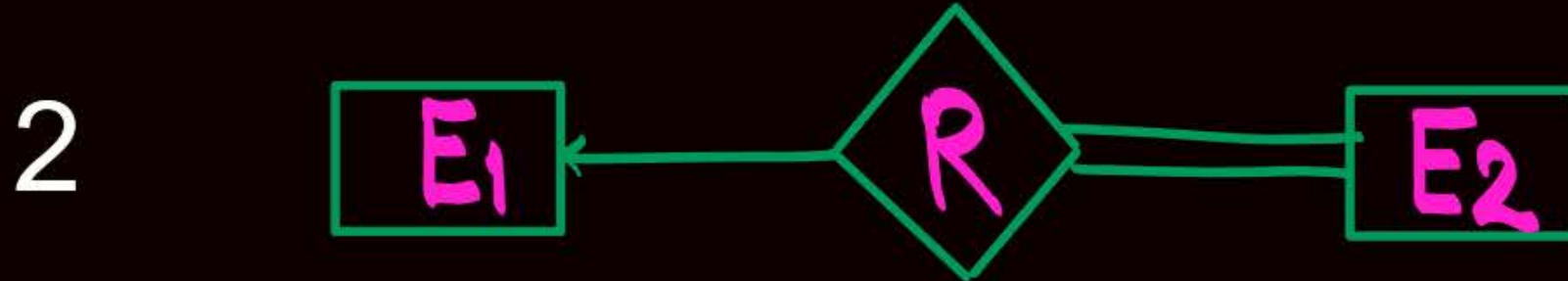
4



{ One table is required
We can choose either P.K. of E1 as
P.K. of Combined relation OR we can choose
P.K. of E2 as P.K. of Combined relation }

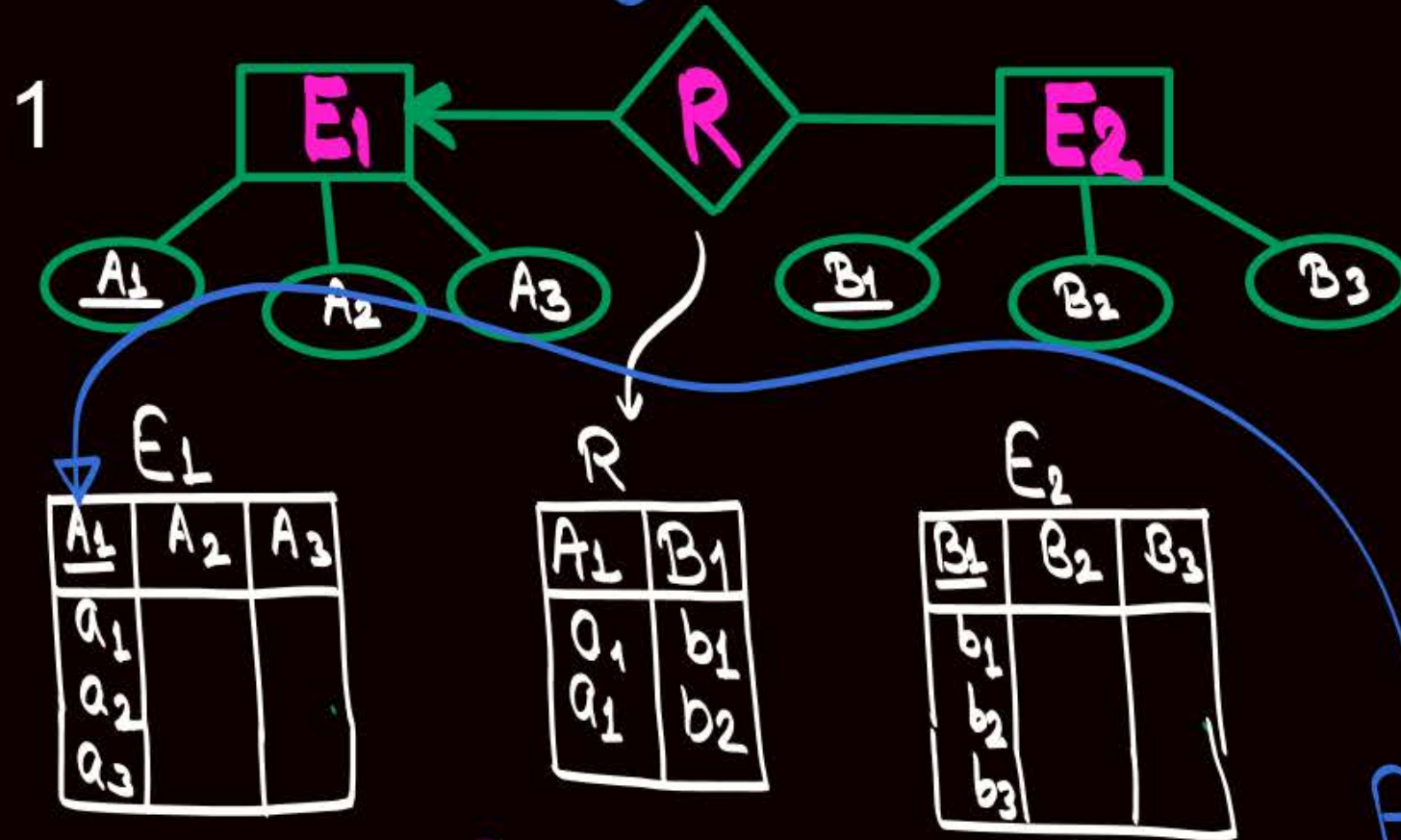
ER Model to Relational Table

② One to Many Relation



ER Model to Relational Table

② One to Many Relation { Partial participation of both side entities }



E1 E2 R

A1	A2	A3	B1	B2	B3
a1	-	-	b1	-	-
a1	-	-	b2	-	-
a2	-	-	NULL	NULL	NULL
a3	-	-	NULL	NULL	NULL
NULL	-	-	b3	-	-

E1 R

duplicate

A1	A2	A3	B1
a1	-	-	b1
a1	-	-	b2
a2	-	-	NULL
a3	-	-	NULL

No P.K. can be defined
 ∴ R cannot merge with E1
 i.e. 1 side entity set

E2 R

<u>B1</u>	B2	B3	A1
b1	-	-	a1
b2	-	-	a1
b3	-	-	NULL

We can merge R with E2 (many side entity set)
 P.K. will be same as many side entity set

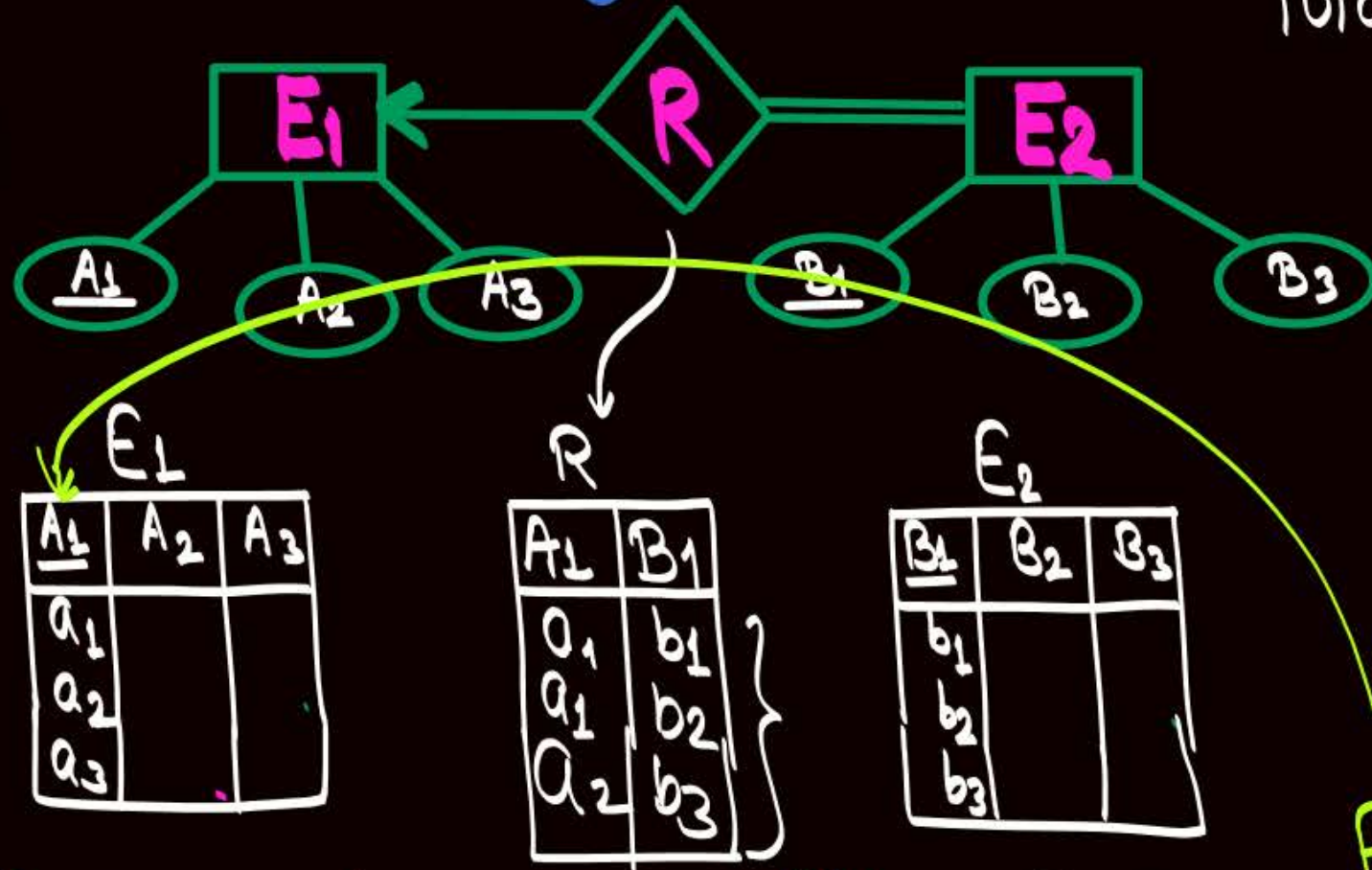
No. primary key can be defined,
 ∴ can not be merged into a single relation

ER Model to Relational Table

② One to Many Relation

{ Partial participation of '1' side entity set,
Total participation of 'Many' side entity set

2



E1

<u>A1</u>	A2	A3
a1		
a2		
a3		

R

A1	B1
a1	b1
a1	b2
a2	b3

E2

<u>B1</u>	B2	B3
b1		
b2		
b3		

E1 R

A1	A2	A3	B1
a1	-	-	b1
a1	-	-	b2
a2	-	-	b3
a3	-	-	NULL

No Primary key
∴ R can not be combined with '1' side entity set

E2 R

<u>B1</u>	B2	B3	A1
b1	-	-	a1
b2	-	-	a1
b3	-	-	a2

Note: All three can not be merged into single relⁿ
 { A1: will have duplicate value
 { B1: will have null values
 ∴ No P.K.

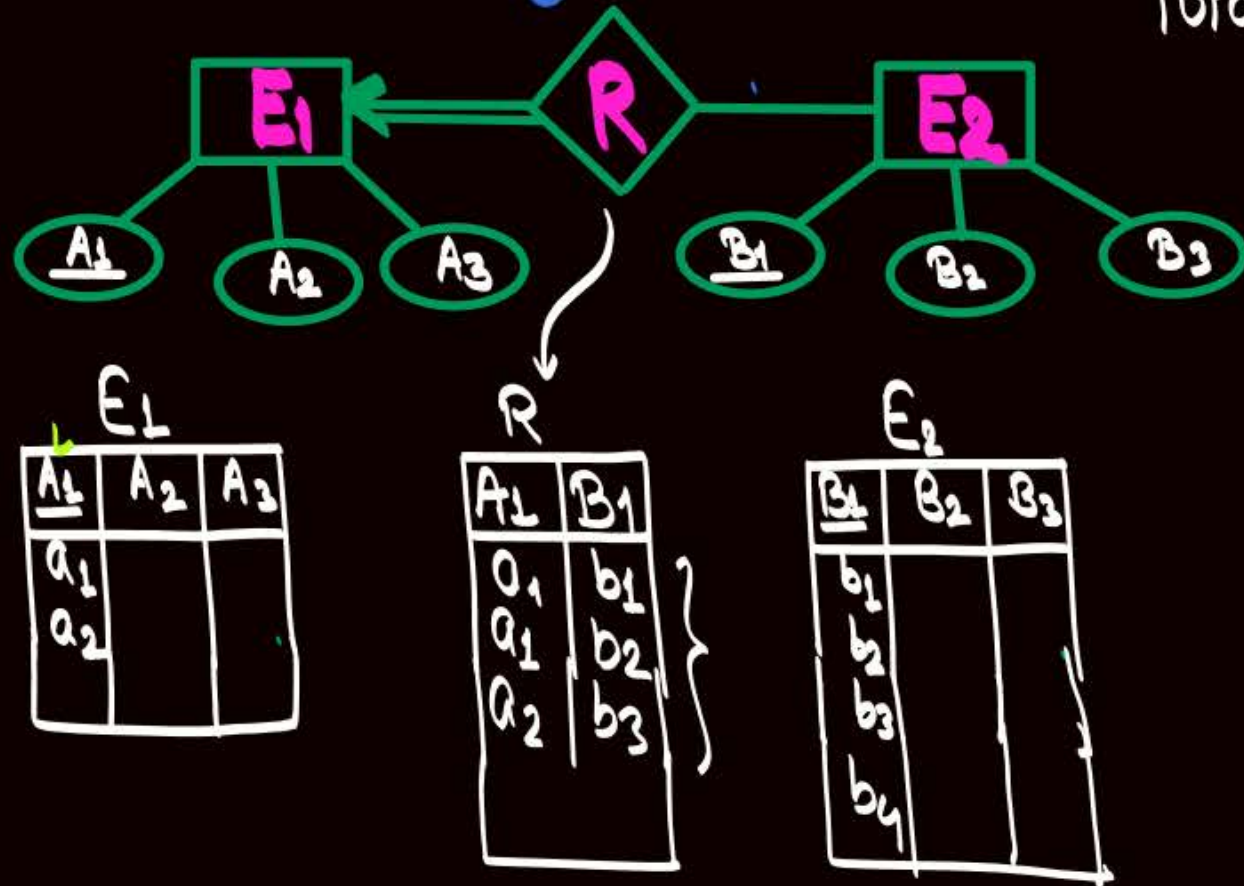
We can merge R with E2 [i.e., many side entity set]
 And primary key of merged relⁿ will be same as P.K. of many side entity set.

ER Model to Relational Table

② One to Many Relation

{ Partial participation of "many" side entity set,
Total Participation of "1" side entity set

3



E1 E2 R

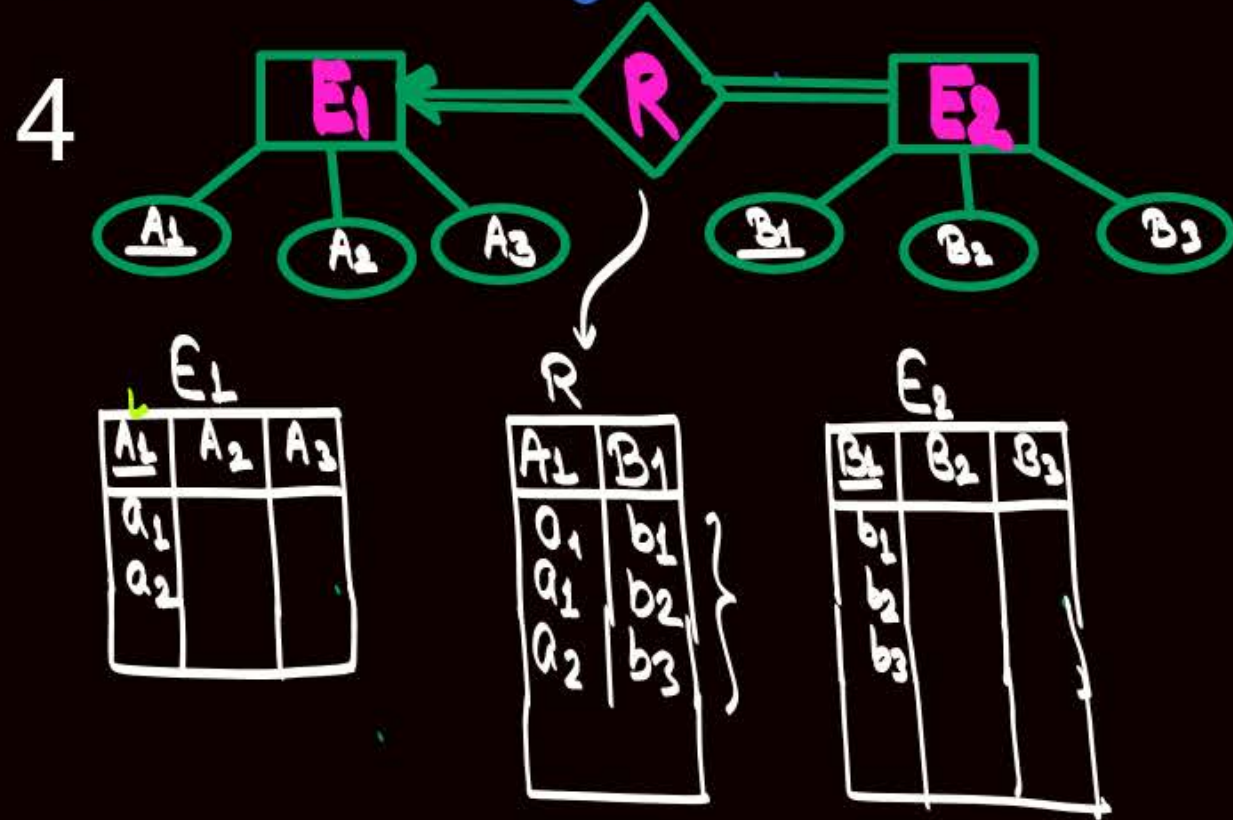
A1	A2	A3	<u>B1</u>	B2	B3
a1	-	-	b1	-	-
a1	-	-	b2	-	-
a2	-	-	b3	-	-
NULL	-	-	b4	-	-

Unique
∴ Primary Key

Hence all three
Can be merged into
a single relation.
And primary key
of merged relⁿ will
be same as P.K.
of many side
entity set.

ER Model to Relational Table

② One to Many Relation { Total participation of both entity set }



$E_1 \ E_2 \ R$

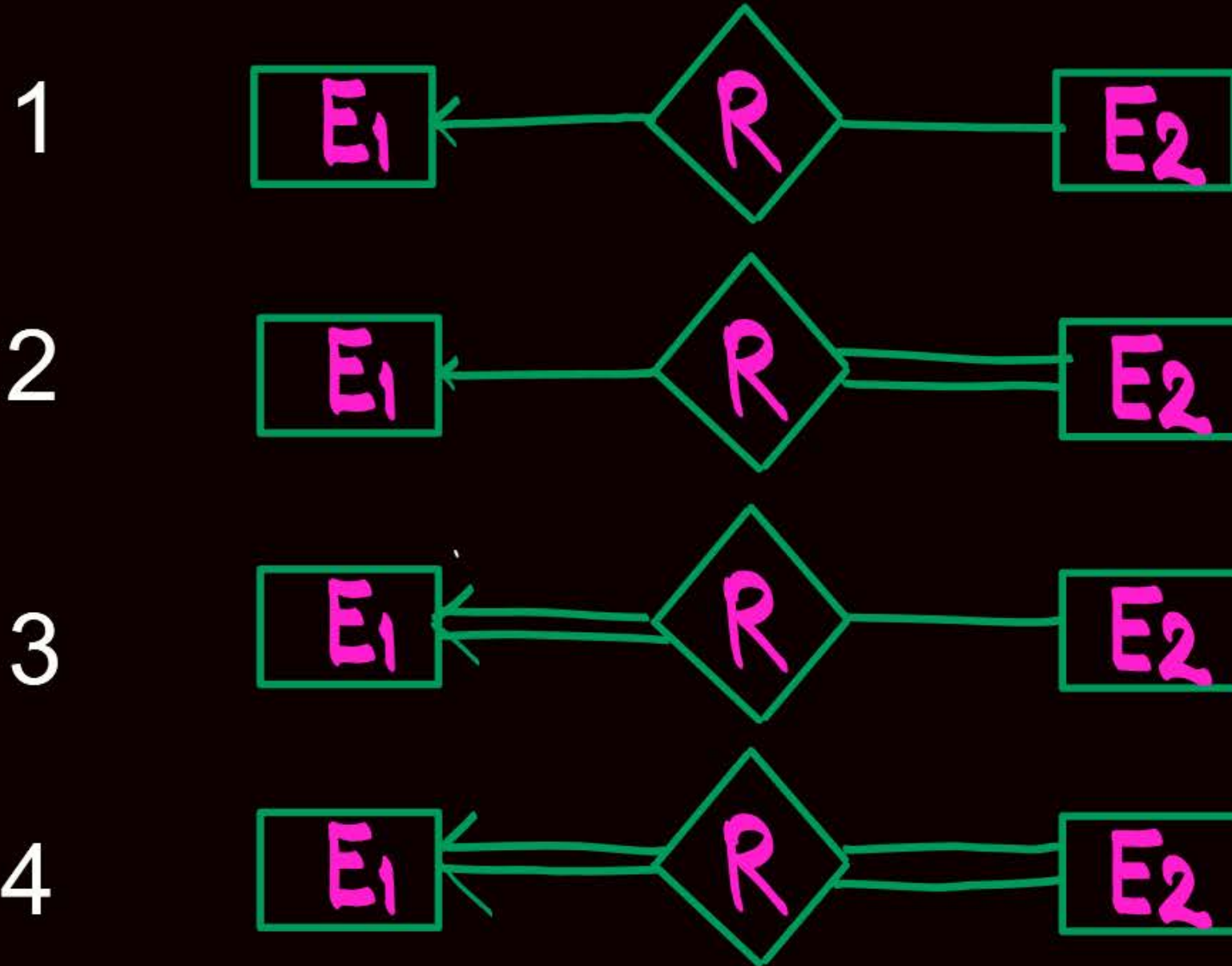
A_1	A_2	A_3	<u>B_1</u>	B_2	B_3
a_1	-	-	b_1	-	-
a_1	-	-	b_2	-	-
a_2	-	-	b_3	-	-
-	-	-	-	-	-

unique
i.e. Primary Key

Hence all three
Can be merged into
Single relⁿ, and
P.K. of merged relⁿ
Will be same as
Primary key of
many side entity set

ER Model to Relational Table

② One to Many Relation



{ Two tables are required }
{ Relation will merge with many side entity set, P.K same as many side entity }

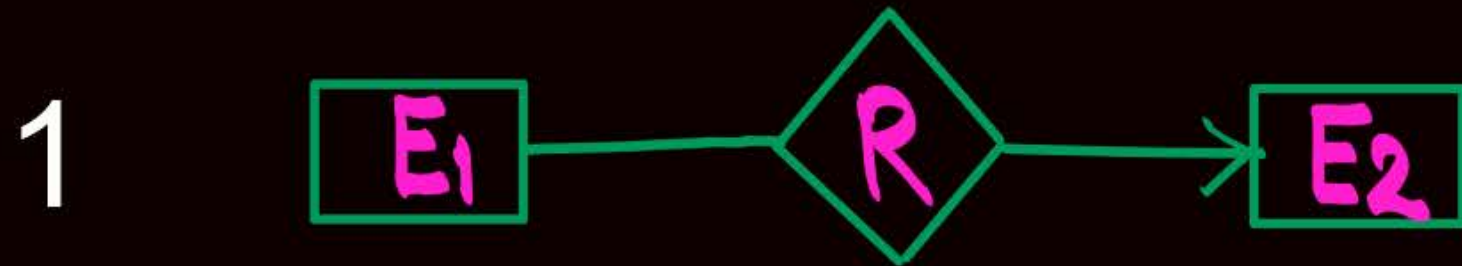
{ Two tables are required }
{ Relation will merge with "many" side entity set, P.K same as many side entity. }

{ only one table is required }
{ P.K. of merged relation will be same as P.K. of many side entity }

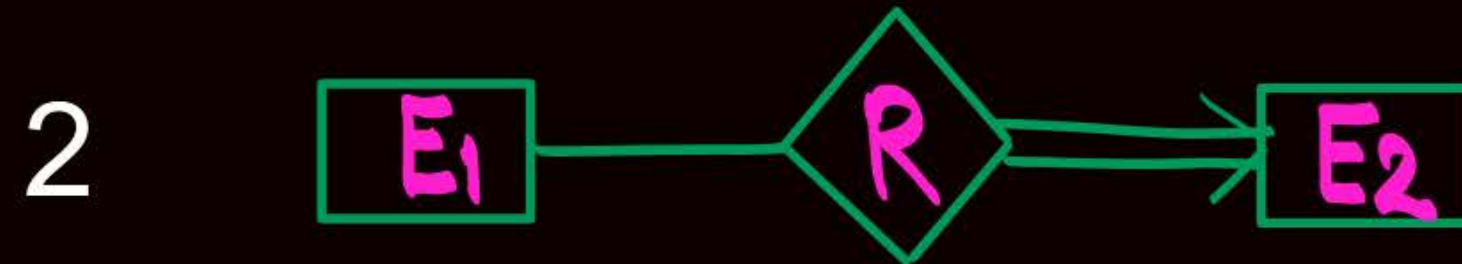
{ only one table is required }
{ P.K. of merged relation will be same as P.K. of many side entity }

ER Model to Relational Table

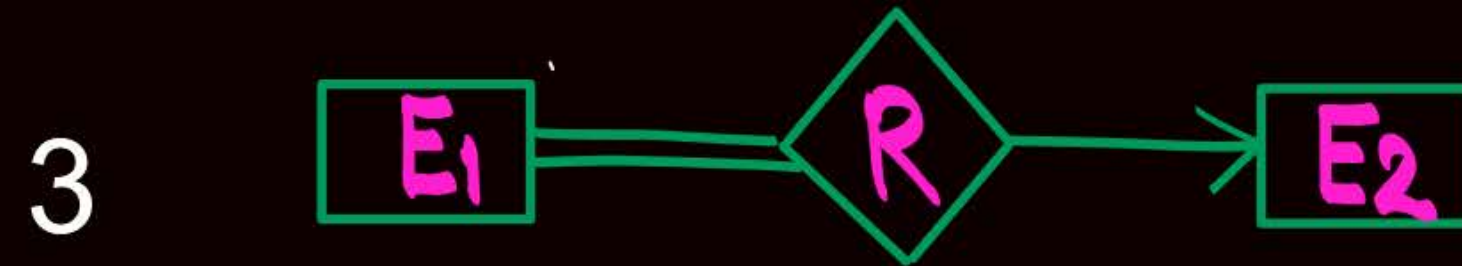
③ Many to One Relation { same concept, as one-to-many }



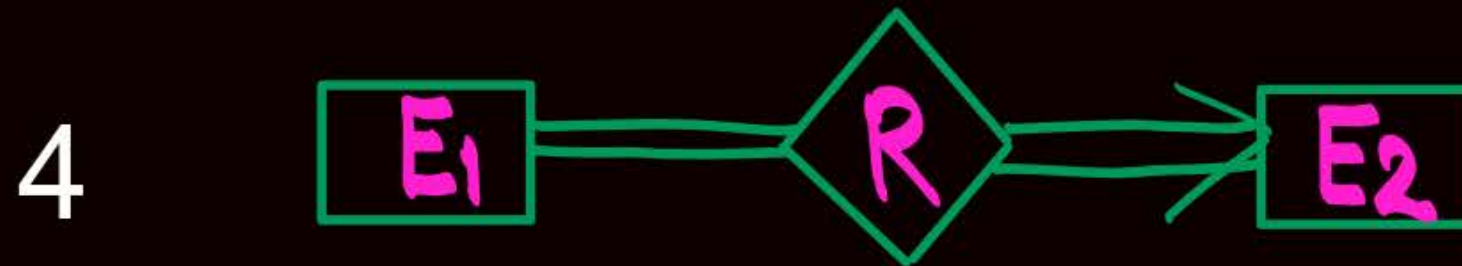
{ Two tables are required
Relation will merge with many side entity set. P.K. of merged relⁿ same as P.K. of many side entity }



{ only one table is required,
P.K. of merged relation will be same as P.K. of many side entity }



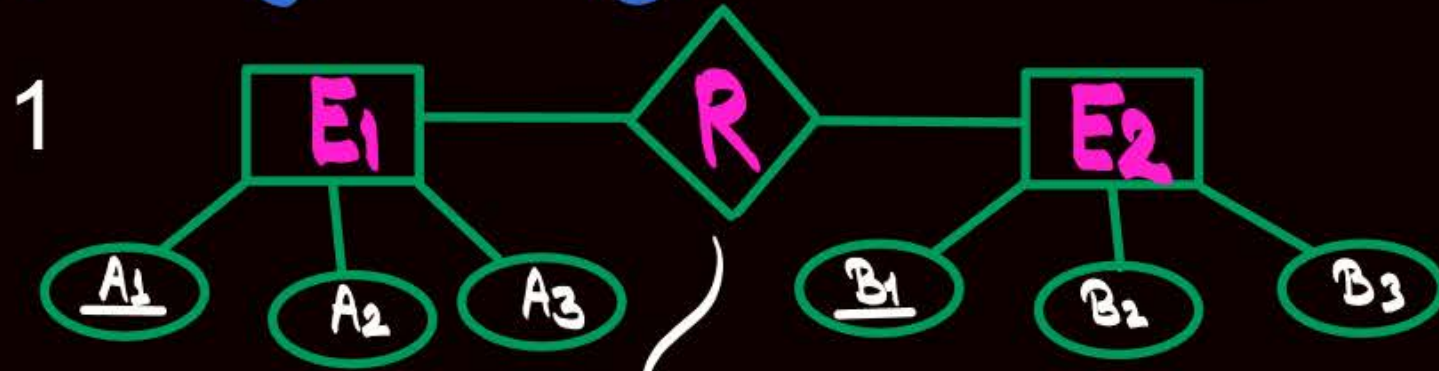
{ Two tables are required
Relation will merge with "many" side entity set. P.K. of merged relⁿ will be same as P.K. of many side entity set }



{ only one table is required,
P.K. of merged relation will be same as P.K. of many side entity set }

ER Model to Relational Table

④ Many to Many Relation { Partial participation of both side entities }



E1

<u>A1</u>	A2	A3
a1		
a2		
a3		

R

<u>A1</u>	<u>B1</u>
a1	b1
a1	b2
a2	b2

E2

<u>B1</u>	B2	B3
b1		
b2		
b3		

E1R

<u>A1</u>	A2	A3	<u>B1</u>
a1	-	-	b1
a1	-	-	b2
a2	-	-	b2
a3	-	-	NULL

No. P.K.
∴ R
Can not be merged with E1

E2R

<u>B1</u>	B2	B3	<u>A1</u>
b1	-	-	a1
b2	-	-	a1
b2	-	-	a2
b3	-	-	NULL

No. P.K.
∴ R Can not be merged with E2

We will require three tables in the relational model,
One - one for each entity set,
One - for relationship set.

ER Model to Relational Table

④ Many to Many Relation { Partial Participation at one side, total participation at another side }

2



All three can not be merged into a single relation

Two tables are required
 $E_2 R(B_1, B_2, B_3, A_1)$ & $E_1(A_1, A_2, A_3)$

<u>A1</u>	A2	A3
a1		
a2		
a3		

<u>A1</u>	B1
a1	b1
a1	b2
a1	b3
a2	b3

<u>B1</u>	B2	B3
b1		
b2		
b3		

A1	A2	A3	B1
a1	-	-	b1
a1	-	-	b2
a1	-	-	b3
a2	-	-	b3
a3	-	-	NULL

No. primary key
 Possible for (E1R)

B1	B2	B3	A1
b1	-	-	a1
b1	-	-	a1
b1	-	-	a1
b2	-	-	a1
b3	-	-	a2

Not unique

tuple wise
 (B1, A1) is unique

{B1, A1} can be defined as P.K. of (E2R)

Note: Relation R can be merged with the entity set with total participation

And P.K. of the merged relation will be defined by combining the P.Ks of both the entity sets.

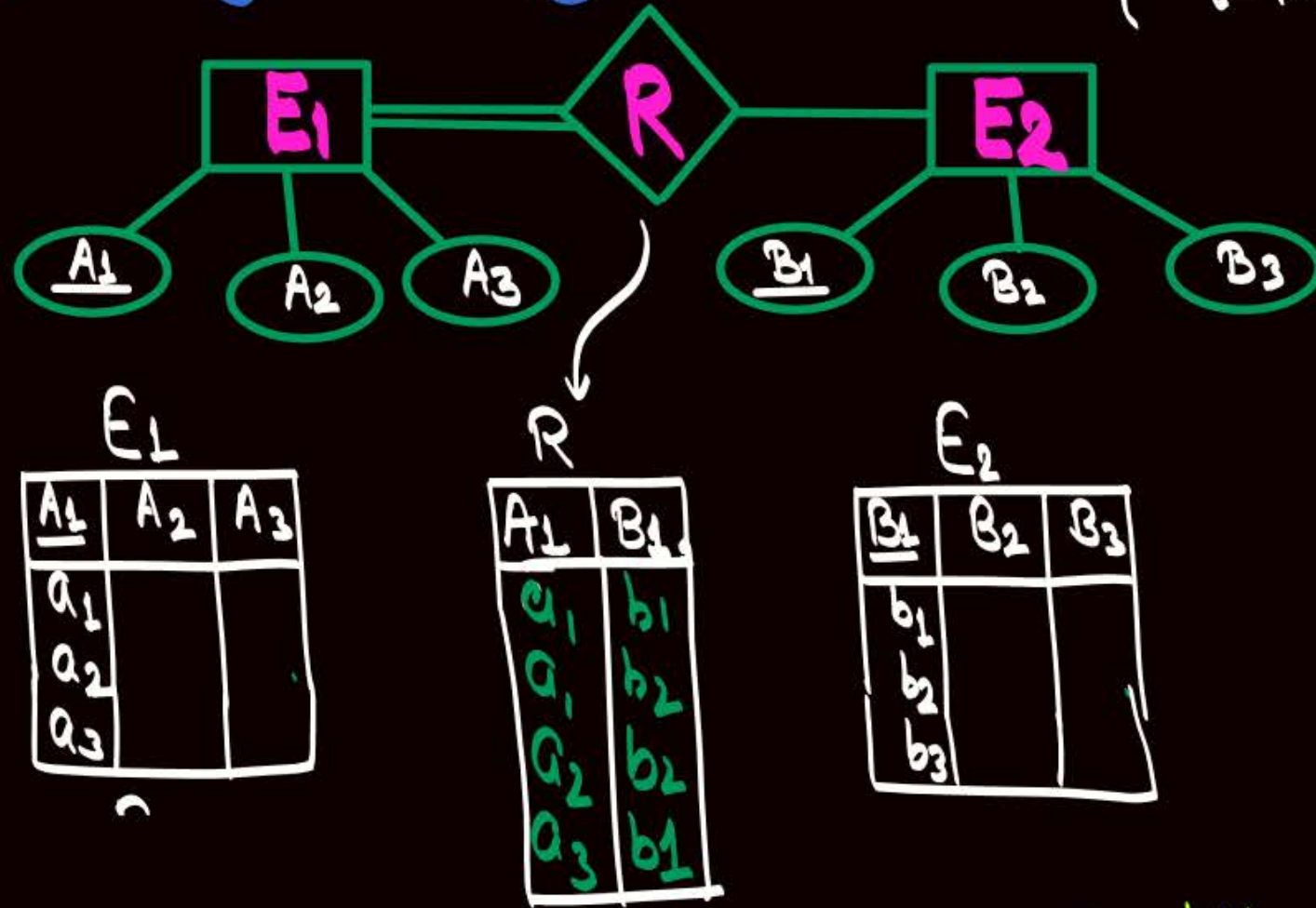
not unique

not unique

ER Model to Relational Table

④ Many to Many Relation { Partial Participation at one side, total participation at another side }

3



Same concept as previous slide,

Two tables are required
(E1R) & E2

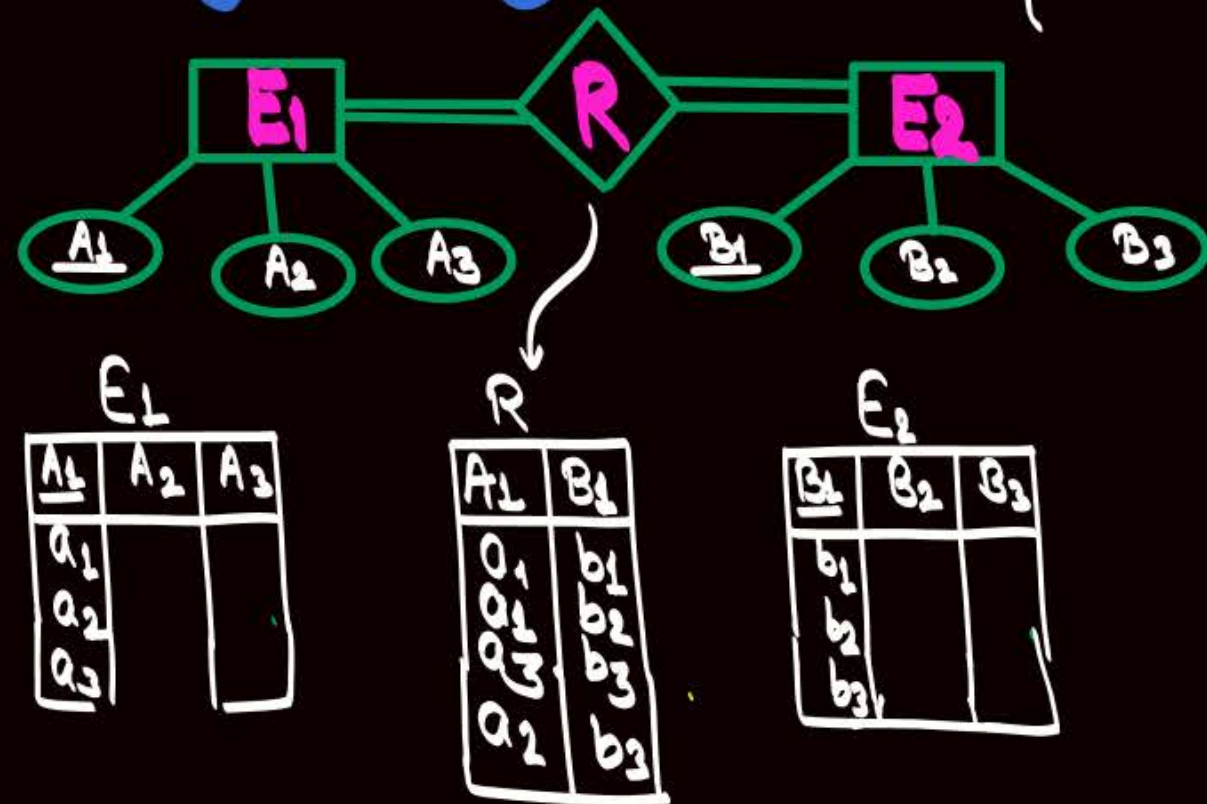
P.K of R = (P.K of E1 + P.K of E2)

Two tables =

{ E1R (A1, A2, A3, B1)
E2 (B1, B2, B3) }

ER Model to Relational Table

④ Many to Many Relation { Total Participation at both side }



$E_1 E_2 R$

<u>A_1</u>	A_2	A_3	<u>B_1</u>	B_2	B_3
a_1	-	-	b_1	-	-
a_2	-	-	b_2	-	-
a_3	-	-	b_3	-	-
a_2	-	-	b_3	-	-

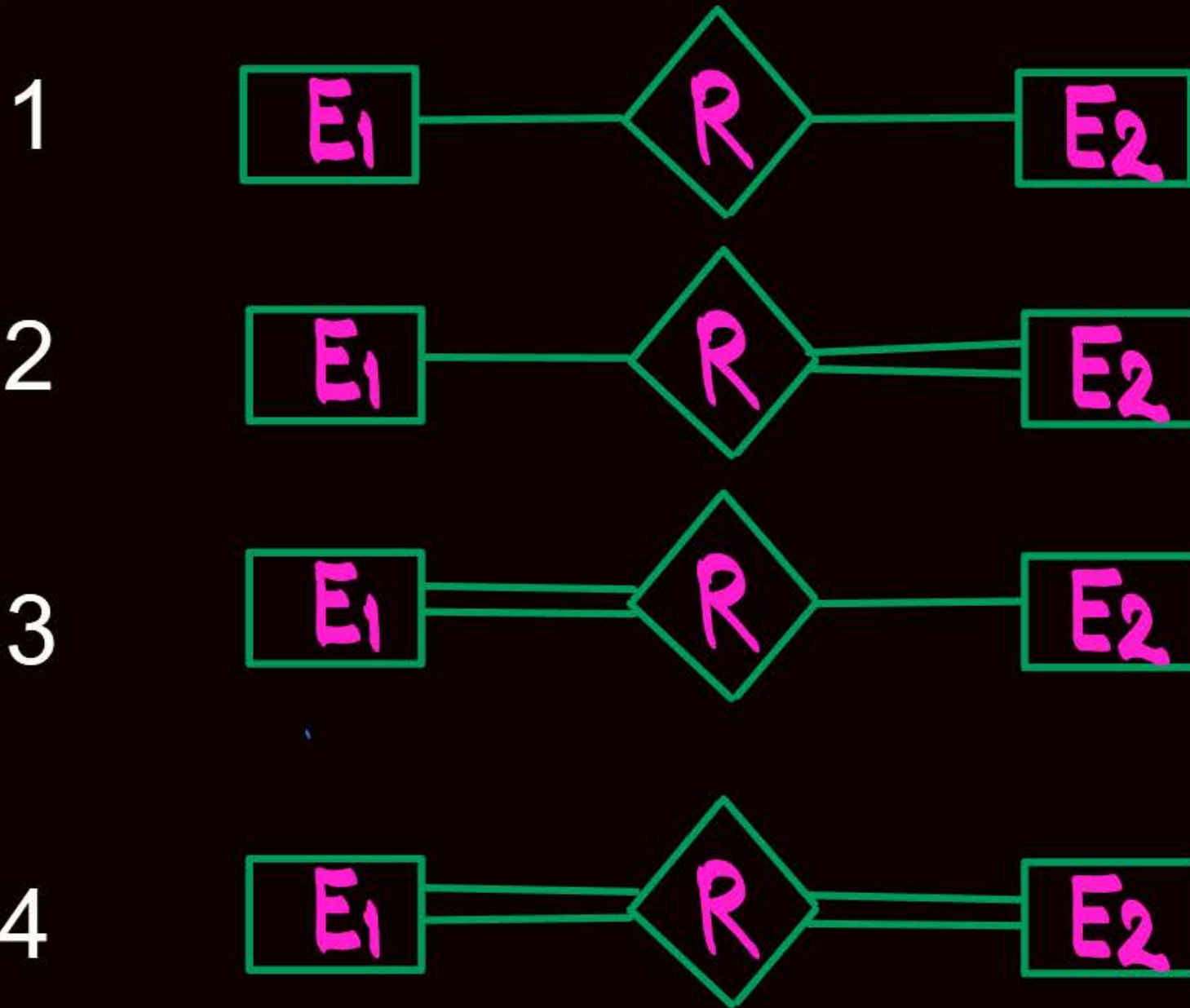
P.K.

All three can be merged into a single table

Primary Key = Primary Key of the merged relation is defined by combining P.K.s of both the entity sets.

ER Model to Relational Table

④ Many to Many Relation



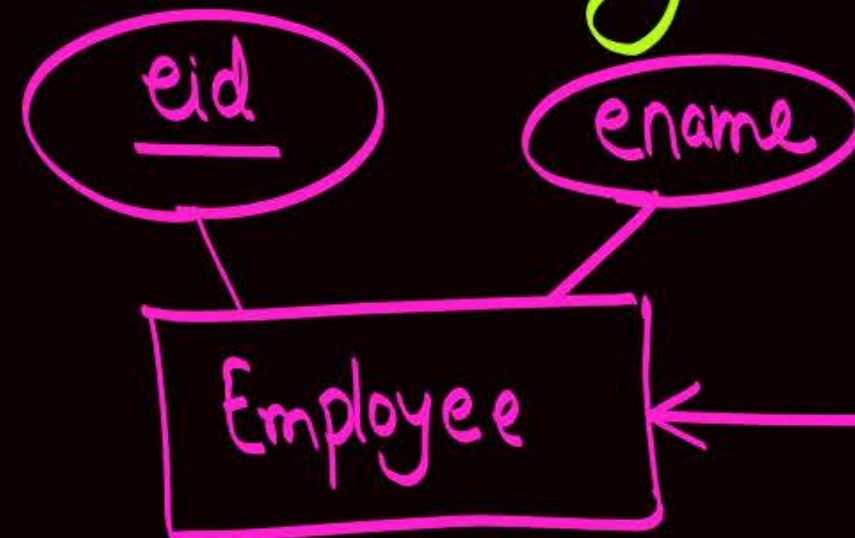
{ minimum three tables are required {No merging of tables} }

{ Two tables are required, relation R will merge with entity set with total participation, and P.K. of merged Relⁿ will be P.K.s of both the entitysets together }

{ Two tables are required, relation R will merge with entity set with total participation, and P.K. of merged Relⁿ will be P.K.s of both the entitysets together }

{ only one table is required. P.K. of merged Relⁿ will be formed by combining P.K.s of both the entitysets }

④ FR model to relational model w.r.t.
Week entity set

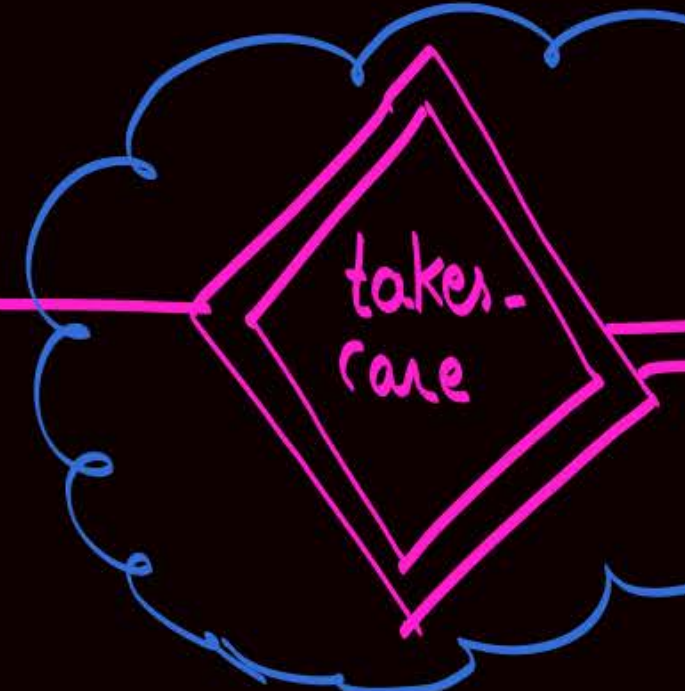


Relation w.r.t. Employee
Entity Set

Primary Key

<u>eid</u>	ename

Employee



Relation w.r.t. dependent
Dependent

PK

<u>eid</u>	<u>d_name</u>	Age



2 mins Summary



✓
Topic

Integrity constraints

✓
Topic

Conversion from ER model to relational table

time/vishal Rawtiya pw



THANK - YOU